Artificial Recall: A Cue-First, Least-Sufficient-Intervention Architecture for Invasive Memory Input/Output

Version 2 — Design Proposal

test-spec-tech

July 26, 2026

Abstract

Version 1 of this project treats memory writing as the objective and patterned electrical stimulation as the mechanism of first resort. We argue that this inverts the correct design order. The deliverable of a memory prosthesis is not a written engram; it is a *reliable, verified, on-demand recall event*. Version 2 reorganizes the architecture around that reframing into three principles — **cue-first** addressing, **least-sufficient intervention**, and **provenance-carrying** output — and introduces six subsystems absent from V1: a Cue Presentation Layer (CPL) that delivers progressively disclosed, modality-agnostic retrieval cues, a Least-Sufficient-Intervention controller (LISI) that escalates through five rungs and stops on verified success, a Confabulation Discrimination Layer (CDL) that separates genuine recall from stimulation-induced false memory, an addressable Artificial Recall Index (ARI), an Interference-Aware Capacity Manager (IACM), and a Cross-Session Aligner (XSA). We give a complete engineering budget: a 1024-channel intracortical read path closes spike-to-cue in 45.3 ms typical, on-implant reduction compresses 307.2 $\mathrm{Mbit\,s^{-1}}$ of broadband data by 750×, and the write chain completes in 11.4 ms typical (28.0 ms worst case), fitting inside the hippocampal sharp-wave ripple window. Every figure in this paper is either computed from a stated formula or drawn from one of **189 independently re-resolved 2024–2026 references**, of which **85 are from 2026**. **This is a design proposal. No component has been built, implanted, or tested, and no experimental result is reported.**

1 Introduction

A memory prosthesis is usually specified as a device that writes information into the brain. That specification is natural and, we argue, wrong — or at least premature. It makes the stimulator the centerpiece, it makes verification a downstream afterthought, and it obliges the system to solve the hardest problem in the stack (constructing a synthetic engram that the subject's own network will accept and later retrieve) before it has solved the easiest one (getting the subject to remember something they already encoded).

The prior version of this project (V1) is a design corpus in the `test-spec-tech/bci-notes` repository. It is not a paper: it contains no data, no code, no figures, and no results, and its own first line advises the reader to treat its content with skepticism because it was drafted with a language model. Read as a source of design intent rather than evidence, V1 is nonetheless substantial: it specifies a six-stage write pipeline, a MIMO/Volterra-Poisson CA3→CA1 framework for generating write codes, manifold-constrained pattern injection, a five-layer convergent-evidence verification structure, and a staged rodent→primate→human experimental programme. It also states plainly that artificial memory writing is invasive-only, which is the premise this version retains.

What V1 does not contain, anywhere in its 2,151 lines, is a presentation layer, a user interaction model, an escalation policy, or an answer to the question it raises against itself: how would you know whether a reported memory was retrieved or manufactured?

1.1 Contributions

1. A reframing of the objective from *engram written* to *recall event verified*, and the resulting cue-first architecture (§2).
2. **LISI**, a five-rung intervention ladder with an explicit stopping rule, which makes minimal stimulation a controller property rather than an aspiration (§3).
3. **CPL**, a modality-agnostic cue-presentation layer that is rung 0 of that ladder — structural, not peripheral (§6).
4. **CDL**, a confabulation-discrimination layer added as Layer 6 of the evidence stack (§7).
5. **ARI**, **IACM**, and **XSA**: addressable memory registry, interference gating, and per-session decoder realignment (§4, §5).
6. A complete, source-traceable latency, bandwidth, and information-rate budget (§8), and a falsifiable validation protocol (§9).

2 The V2 thesis

If what the system owes the user is a recall event, three properties follow directly.

Cue-first. The brain addresses memories by partial cue into an attractor network, not by pattern injection. Human intracranial recordings show that context and stimulus identity are decodable from medial-temporal population activity well above chance [Bausch et al., 2026], and that hippocampal ripples initiate cortical dimensionality expansion during retrieval [Kerrén et al., 2026] and coordinate planning sequences and compositional representations [He et al., 2026]. A system that can *address* that machinery does not need to replace it.

Least-sufficient intervention. Every rung of intervention carries cost: charge delivered to tissue, interference with existing traces, and risk of manufacturing content. The correct policy is therefore to spend the smallest intervention that produces the recall and stop immediately — which requires a verification signal in the loop, not at the end of it.

Provenance-carrying. Every recall event the system surfaces must be labelled with where it came from: endogenous retrieval, cued retrieval, or stimulation-assisted reinstatement. A prosthesis that cannot tell the user which of these occurred is not safe to use, regardless of its accuracy.

3 The Recall Ladder

LISI walks five rungs in strict order of increasing invasiveness and terminates on the first verified recall (Fig. 3, left).

Rung	Intervention	Cost
0	Cue presentation (CPL)	none
1	Endogenous retrieval monitoring	none
2	Sub-threshold entorhinal bias	minimal charge
3	Partial-seed CA3 injection	moderate
4	Full patterned MIMO reinstatement	maximum

Table 1: The intervention ladder. V1 contemplated only rungs 3 and 4.

The stopping rule is what gives the ladder force. LISI polls the CDL after every rung, so the expected charge delivered is dominated by the cheapest rung that works rather than by the worst case. This matters because stimulation of medial-temporal structures has heterogeneous and sometimes negative effects on memory: direct brain stimulation outcomes depend on structural and functional connectivity [Zhang et al., 2026a], continuous bipolar hippocampal DBS in temporal-lobe epilepsy patients has measurable cognitive consequences [Ortiz et al., 2026], and fornix stimulation for Alzheimer's disease has a mixed evidence base [Fahim et al., 2026]. An architecture whose default is maximum stimulation inherits all of that risk unnecessarily.

4 Read path

Sensing and on-implant reduction. The design point is 1024 channels across CA1, CA3, entorhinal cortex, and prefrontal cortex, sampled broadband at 30 kHz and 10-bit resolution. This is a conservative channel count by 2026 standards: a recording and stimulation platform with 5,376 simultaneous channels at 20 kS/s has been reported, producing in excess of $1.3\,\mathrm{Gbit\,s^{-1}}$ of throughput [Fan et al., 2026]. Our configuration produces $307.2\,\mathrm{Mbit\,s^{-1}}$, which no fully implanted wireless device can transmit; the same constraint drives contemporary headstage designs toward adaptive sampling and on-board reduction [Liu et al., 2026], and is more severe still for fully wireless endovascular embodiments, where power-transfer efficiency falls to a few percent at clinically relevant depths [Tai et al., 2026]. Spike detection and 20 ms binning therefore run on-implant, reducing the uplink to $0.41\,\mathrm{Mbit\,s^{-1}}$ — a $750\times$ reduction (Fig. 3, right). Ripple detection also runs on-implant and emits event tokens, because the write path's timing budget cannot absorb a telemetry round trip (§5).

Cross-session alignment (XSA). Electrode drift and unit turnover mean a decoder trained on one session is not valid on a later one without realignment; this is a first-order obstacle to any chronic memory interface, and V1 simply assumes it away. The 2024–2026 decoding literature converges on the same remedy — learned alignment onto a shared latent space. Recent work demonstrates zero-shot decoding from conserved kinematic representations [Ravishankar and de Sa, 2026], cross-brain transfer of intracortical speech and handwriting decoders [Levin et al., 2026], shared latent representations supporting cross-patient speech decoding [Wadia et al., 2026], adversarial alignment for generalizable speech neuroprostheses [Zhang et al., 2026b], and unsupervised use of unlabelled data for generalizable population decoding [Mao et al., 2026]. XSA fits a linear map onto a stored reference manifold before each session's decoding begins. Hardware-side mitigation is complementary rather than sufficient: thin-film arrays achieving low-drift, repeatable recordings in behaving primates still report a single-unit success rate below unity [Woods et al., 2026], so a software realignment stage remains necessary.

Decoding to an address, not to content. The decoder emits a distribution over ARI registry addresses rather than a reconstruction of remembered content. This is a deliberate restriction: the content is whatever the subject's own network reinstates at that address, which keeps the system on the side of *addressing* memory rather than *authoring* it, and sharply narrows the privacy surface.

Verification (CDL). A separate lightweight classi-

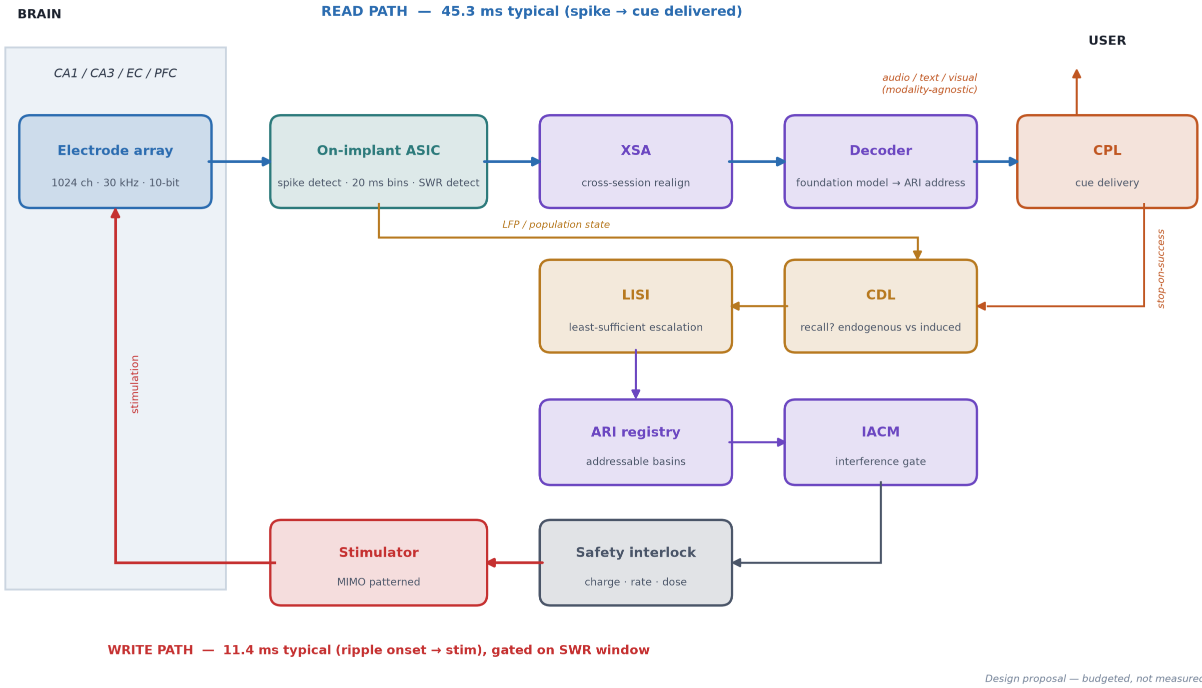

Figure 1: Artificial Recall V2 system architecture. The read path (blue) runs from a 1024-channel array through on-implant reduction, cross-session alignment (XSA), and a foundation-model decoder to the Cue Presentation Layer. The verification path (amber) — CDL and LISI — is a closed loop, not a terminal stage: it gates escalation and supplies the stopping signal. The write path (red) is gated three times, by the ripple window, by manifold projection, and by the IACM, before a hardware safety interlock releases any stimulus. Latency figures are computed budgets (Table 3), not measurements.

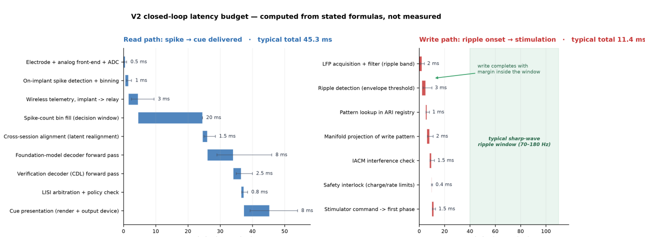

Figure 2: Closed-loop latency budget. Read path (left) and write path (right), stage by stage, with min/max whiskers. The write chain's worst case (28.0 ms) fits inside a hippocampal ripple event. All values are computed from the formulas in Table 3; none is measured.

fier answers a different question from the decoder: did a recall occur, and was it endogenous or induced? The CDL supplies LISI's stopping signal and the provenance label on every output.

5 Write path

Writing is gated three times, and all three gates must pass.

1. The ripple window. Writes are attempted only during detected sharp-wave ripples, the physiological window for hippocampal–cortical transfer. Human hippocampal ripples are reported in the 70 Hz to 180 Hz band [Zhang et al., 2025] with a peak frequency near 94.7 Hz over 5,404 detected events [Huang et al., 2025], occur at roughly 787 events per participant per session [Kerrén et al., 2026], and are present during the large majority of trials [Huang et al., 2025]. They are modulated by arousal state [Siefert et al., 2026] and tune cortical responses according to predicted uncertainty [Frank et al., 2026]. The full detection-to-stimulation chain is budgeted at 11.4 ms typical precisely so that it fits inside such an event.

2. The manifold. A pattern lying off the population's intrinsic manifold is rejected by recurrent dynamics and does not become a retrievable memory. The write pattern is projected onto the manifold and its residual checked against V1's $\mu + 2\sigma$ orthogonality threshold.

3. Capacity (IACM). The new pattern must land far enough from every existing basin in the ARI. If it does not, the IACM refuses the write or re-codes it to a free address. V1 names interference as a risk but supplies no mechanism; the IACM is that mechanism.

Partial-seed writing. V2 retains V1's proposal to inject a seed and rely on CA3 pattern completion, but explicitly declines to state a seed fraction. V1's "20–30%" figure is labelled in V1's own text as unmeasured, and we do not propagate it. Determining the seed fraction is an experimental question (§9).

Safety interlock. A hardware check on charge density, charge per phase, pulse rate, and cumulative daily

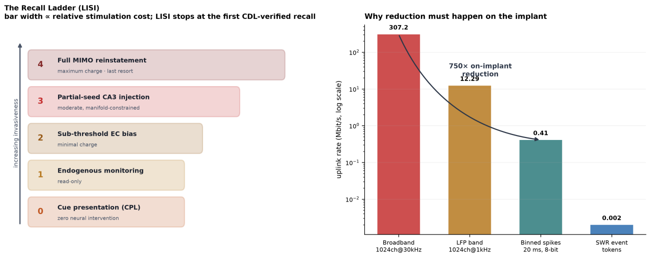

Figure 3: Left: the intervention ladder, bar width proportional to relative stimulation cost; LISI stops at the first CDL-verified recall. **Right:** why data reduction must happen on the implant — broadband telemetry exceeds any implantable wireless budget by orders of magnitude.

dose sits between the pattern generator and the stimulator and can veto any command. It is deliberately implemented outside the decoder's software domain. That chronic intracortical microstimulation can be delivered safely at scale is now supported by human data: a five-participant series reports more than 168 million stimulation pulses delivered over a combined 27 years of implant duration without serious adverse events [Greenspon et al., 2026]. This is an argument for a dose-bounded architecture, not against one — the interlock is what makes such a record auditable.

6 Cue Presentation Layer

The CPL is rung 0. It delivers cue elements — contextual, semantic, temporal — reconstructed from the encoding context stored in the ARI, and discloses them progressively: least specific first, one element at a time, halting the instant the CDL reports hippocampal state completion. The design objective is the *minimum sufficient cue*.

The layer is **modality-agnostic by design**. A cue may be delivered as audio, as text, as an image, or as a contextual prompt on whatever device the subject already uses; no part of the architecture depends on a particular output device, and the latency budget accordingly carries a single generic presentation stage (2 ms to 16.7 ms, spanning audio playback through one frame of a 60 Hz visual display) rather than a display pipeline. This matters for feasibility: rung 0 must be the cheapest tier to deploy as well as the least invasive, and binding it to specialized hardware would defeat that.

Two properties distinguish the CPL from a passive output. First, stop-on-success makes the presentation loop closed: disclosure rate is driven by the verification decoder, not by a fixed script. Second, **arbitration**: when decoded content and user-confirmed content disagree, the CPL surfaces the disagreement rather than silently resolving it. Decoded content is never presented as fact without its provenance label.

7 Evidence stack

Layer	Evidence	Status
1	Behavioral recall accuracy	prior art
2	Neural reinstatement	prior art
3	Content specificity	prior art
4	Temporal specificity	prior art
5	Dose–response	prior art
6	**Confabulation discrimination**	**new**

Table 2: Layer 6 (CDL) is V2's central scientific addition.

Layers 1–5 are, jointly, insufficient. A subject may report a vivid memory, exhibit matching neural reinstatement, locked in time to the intervention and scaling with its dose, and still be reporting a memory the system created rather than one it retrieved. Every one of the first five layers is satisfied in that scenario. This is not a hypothetical concern for a device whose rung-4 behaviour is to inject a pattern designed to be accepted by the subject's network as a memory.

The CDL targets that failure mode by learning the signature that distinguishes endogenous retrieval dynamics from driven reinstatement. We state plainly that this is the least specified component in the architecture: we pose it as a requirement and a validation gate, not as a solved problem. Its plausibility rests on the observation that endogenous retrieval has characteristic large-scale dynamics — ripple-initiated cortical dimensionality expansion [Kerrén et al., 2026], rapid hippocampal–neocortical interplay [Wang et al., 2026], and distinguishable hippocampal response profiles during verbal memory processing [Huang et al., 2026] — that a driven pattern need not reproduce.

8 Quantitative budget

Quantity	Value	Unit
Channels	1024	ch
Broadband uplink	307.2	Mbit/s
Binned-spike uplink	0.41	Mbit/s
On-implant reduction	750	×
Read latency, typical	45.3	ms
Read latency, range	28.4–79.7	ms
Write latency, typical	11.4	ms
Write latency, worst case	28.0	ms
Ripple band (literature)	70–180	Hz
Ripple peak (literature)	94.7	Hz

Table 3: Budget summary. Full derivations, per-stage values, and per-row source attribution are in the accompanying `v2_budget.csv`. Rows are marked *derived*, *literature*, or *assumption*; no row is unattributed.

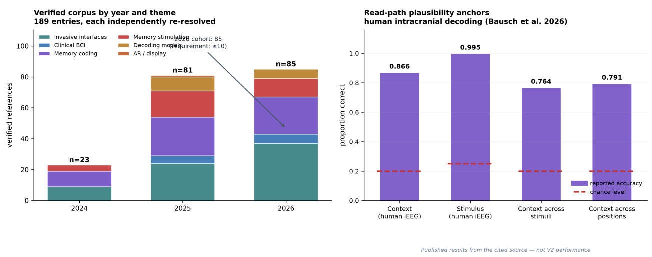

Figure 4: Left: the verified reference corpus by year and theme; every entry was independently re-resolved against Crossref, the arXiv API, or PubMed ID conversion. **Right:** human intracranial decoding accuracies from cited work, with chance levels. These are published results from the cited sources, not V2 performance.

The two figures that matter are these. The read path closes in under 50 ms typical, which is inside the range where a delivered cue feels contemporaneous with intent. The write path's worst case is 28.0 ms against a ripple event on the order of 100 ms, which is the margin that makes ripple-gated writing feasible at all — and it is only achievable because ripple detection sits on the implant rather than across the telemetry link.

9 Validation protocol

Falsifiable core claim. *Least-sufficient-intervention escalation achieves a target recall probability at strictly lower cumulative stimulation charge than fixed full-pattern stimulation, without loss of content fidelity.* The claim is falsified if, in a within-subject design, LISI's cumulative charge is not lower, or if content fidelity at matched recall probability is lower than in the fixed-stimulation arm.

Staged protocol.

1. **Rodent.** Ripple-triggered partial-seed injection versus full-pattern; primary outcome is charge at matched recall.
2. **Non-human primate.** Behnke–Fried-geometry place-preference writing; adds the manifold and IACM gates.
3. **Human, read-only.** Patients already implanted for clinical indications; CPL rung 0 and CDL validation with **no stimulation**. This stage validates the verification decoder before any human write is attempted.
4. **Human, cued.** Rungs 0–2 only, sub-threshold.
5. **Human, seeded.** Rungs 3–4, only if all prior stages clear their pre-registered thresholds.

Pre-registered thresholds. Content fidelity ≥ 0.80; interference ≤ 0.05; replay fidelity ≥ 0.70; and, new in V2, CDL discrimination AUC ≥ 0.85 before any human write is permitted.

10 Ethics and risk

Risk	Mechanism in V2
Confabulation read as recall	CDL + provenance labelling
Interference with existing memory	IACM capacity gate
Off-manifold injection	manifold projection gate
Charge-density damage	hardware interlock
Decoder drift, wrong address	XSA + CDL backstop
Involuntary readout	user-initiated reads; registry stores addresses
Cognitive integrity	confirmation before write; CPL arbitration
Coercion	*unsolved* — policy, not engineering

Table 4: Risk register. The final row is stated as unsolved.

We flag the last row deliberately. A device that can write to memory and be operated by a third party is a device that can be misused in ways no interlock addresses. We do not claim an engineering solution.

11 Limitations

- **Nothing here is built.** Every performance figure is a budget or a literature anchor. No integrated system exists.
- **No in-house invasive data.** This project's only neural data is non-invasive scalp EEG and MEG recorded under a *keystroke-decoding* paradigm — not a memory paradigm. It cannot support any claim about memory decoding, and we make none.
- **The CDL is underspecified.** It is the most important component and the least solved.
- **Key parameters are unmeasured**: seed fraction, ripple-window tolerance, and basin-separation distance.
- **V1's risk budget is priors, not probability.** The five-factor product V1 reports is a structured prior; it is not a probability of success and is not reproduced here as one.
- **Inherited-number hygiene.** Three V1 figures are excluded by design: the "10–100× consolidation" claim (retracted in V1's own later text), and the "5–15" and "8–15 percentage point" effect estimates (V1's expectations, not measurements).

12 Conclusion

V2's argument is that the hard problem of a memory prosthesis is not writing — it is knowing. Knowing whether a recall happened, knowing whether it was real, and knowing when to stop intervening. Reorganizing the architecture around a verified recall event, rather than around a delivered stimulus, converts three of V1's open risks into components with defined interfaces: LISI makes minimality a controller property, the IACM makes interference a gate, and the CDL makes provenance a measurement. The cue-presentation layer, which V1 lacks entirely, is what makes rung 0 possible and therefore what makes the whole ladder start from zero intervention.

None of this is evidence. It is a specification with a budget, a falsifiable core claim, and an explicit list of what would have to be measured before any of it touches a person.

Reproducibility. The 189-reference corpus, with per-entry identifiers and verification method, is provided as `literature_v2.csv/.json`; the budget model as `v2_budget.csv/.json`; the architecture specification as `v2_architecture.md`. Entries that failed live identifier resolution were dropped, not repaired.

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